

KERALA DIPLOMA CURRICULUM REVISION 2026

Course 5423B

5 Credits: 4

Program Elective

Source-grounded

Spices and Condiments

Study the official course objectives in the source syllabus.

OFFICIAL SOURCE DECLARATION

How this lesson was prepared

The supplied official SITTTTR syllabus PDF for Course 5423B is the authoritative source for course information, objectives, course outcomes, CO-PO mapping, modules, topics, activities, assessment structure and references. Explanations, study prompts and Malayalam support are educational elaboration and are not presented as additional official syllabus text.

Source file: 5423B.pdf from the uploaded official SITTTTR syllabus archive.

Course code and title: preserved from the source.

COURSE DASHBOARD

Official course information

Course information

Item	Official information
Programme	Diploma in Food Processing Technology
Course code	5423B
Course title	Spices and Condiments
Semester	5 Credits: 4
Course category	Program Elective
Periods per week	4(L: 4T:0 P: 0) Periods per semester: 60
Periods per semester	60
Credits	4
Prerequisite	See the official syllabus PDF
Assessment	Use the official assessment section reproduced below

Modules

4 official module sections detected.

Topic entries

12 source-aligned topic entries retained.

Study mode

Theory and application emphasis.

STUDY METHOD**How to use this lesson****First pass**

Read the official source declaration, dashboard and module transcript. Mark unfamiliar terms without changing the official terminology.

Second pass

Use each topic card to build a definition, principle, steps or components, application and exam answer. Attempt the Q&A before opening the answers.

Practical evidence

Where the course is laboratory or workshop based, maintain an observation notebook with procedure, instruments, readings, safety points and conclusion.

Revision pass

Return to the source excerpt, coverage table, activities and model paper. The generated paper is practice material, not an official examination paper.

LEARNING OUTCOMES**Objectives, outcomes and mapping****Course objectives**

- Study the official course objectives in the source syllabus.

Course outcomes

CO	Official outcome	Study level
CO1	sources, production, selection criteria, and 10 Understanding classification based on origin and physical characteristics. Explain the occurrence, process flow sheet, and	See official syllabus
CO2	mechanisms involved in the manufacturing of 15 Understanding coffee powder and instant coffee. Explain the occurrence, process flow, and manufacturing mechanisms of tea, along with the	See official syllabus
CO3	15 Understanding characteristics of different types of tea such as white, green, oolong, and black. Describe the occurrence, processing mechanisms (primary and secondary), and quality control	See official syllabus
CO4	procedures involved in cocoa and chocolate 20 Understanding production.	See official syllabus

CO-PO mapping evidence

Row	Extracted official mapping
CO1	CO1 2
CO2	CO2 2
CO3	CO3 2
CO4	CO4 2

COVERAGE AUDIT

Complete syllabus coverage

Every detected official module is represented below. Each module contains topic cards and an expandable official syllabus transcript to preserve source terminology.

Module coverage

Module	Official heading	Topic entries	Hours
Module 1	classification based on origin and physical characteristics.	3	As specified
Module 2	Explain the occurrence, process flow sheet, and mechanisms involved in the	3	As specified
Module 3	Explain the occurrence, process flow, and manufacturing mechanisms of	3	As specified
Module 4	and quality control procedures involved in cocoa and chocolate production.	3	As specified

MODULE 1

classification based on origin and physical characteristics.

This module is presented from the official syllabus scope for Spices and Condiments. Use the topic cards for learning and the source transcript for exact coverage.

As specified
official hours

CO-linked
see outcomes

Learning goals

- Explain the official Module 1 scope and its relationship to the course outcomes.
- Recognise the terminology, sequence, components and evidence expected in examination answers.
- Connect at least one official topic to a practical, industrial, laboratory or design context.

Key facts

- Official module title: classification based on origin and physical characteristics.
- Official duration: As specified
- Official topic entries captured: 3
- The full source excerpt is preserved below for coverage checking.

▼ Explain Status and scope of spice and flavor 4 Understanding processing industries in India Discuss sources, production, selection criteria of

Explain Status and scope of spice and flavor 4 Understanding processing industries in India Discuss sources, production, selection criteria of is an official topic in Module 1 of Spices and Condiments. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify Explain Status and scope of spice and flavor 4 Understanding processing industries in India Discuss sources, production, selection criteria of using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, explain status and scope of spice and flavor 4 understanding processing industries in india discuss sources, production, selection criteria of should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What Explain Status and scope of spice and flavor 4 Understanding processing industries in India Discuss sources, production, selection criteria of means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 1- Explain Status and scope of spice and flavor 4 Understanding processing industries in India Discuss sources, production, selection criteria of definition/meaning . steps, application . Technical terms English- .

▼ Spices 3 Understanding Classify spices on the basis of origin, physical

Spices 3 Understanding Classify spices on the basis of origin, physical is an official topic in Module 1 of Spices and Condiments. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify Spices 3 Understanding Classify spices on the basis of origin, physical using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, spices 3 understanding classify spices on the basis of origin, physical should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
--------	-----------------

Aspect	What to prepare
Definition/identity	What Spices 3 Understanding Classify spices on the basis of origin, physical means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 1- Spices 3 Understanding Classify spices on the basis of origin, physical definition/meaning. steps, application. Technical terms English-.

▼ 3 Understanding characteristic sources, production, selection criteria, classification on the basis of origin, physical characteristic

3 Understanding characteristic sources, production, selection criteria, classification on the basis of origin, physical characteristic is an official topic in Module 1 of Spices and Condiments. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify 3 Understanding characteristic sources, production, selection criteria, classification on the basis of origin, physical characteristic using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, 3 understanding characteristic sources, production, selection criteria, classification on the basis of origin, physical characteristic should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What 3 Understanding characteristic sources, production, selection criteria, classification on the basis of origin, physical characteristic means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 1-3 Understanding characteristic sources, production, selection criteria, classification on the basis of origin, physical characteristic definition/meaning. steps, application. Technical terms English-

► **Official syllabus detail and terminology (expand in digital version)**

Module study routine: Read the source wording, make a one-page concept map, practise one short answer and one structured answer, then review the Malayalam support notes.

MODULE 2

Explain the occurrence, process flow sheet, and mechanisms involved in the

This module is presented from the official syllabus scope for Spices and Condiments. Use the topic cards for learning and the source transcript for exact coverage.

As specified
official hours

CO-linked
see outcomes

Learning goals

- Explain the official Module 2 scope and its relationship to the course outcomes.
- Recognise the terminology, sequence, components and evidence expected in examination answers.
- Connect at least one official topic to a practical, industrial, laboratory or design context.

Key facts

- Official module title: Explain the occurrence, process flow sheet, and mechanisms involved in the
- Official duration: As specified
- Official topic entries captured: 3
- The full source excerpt is preserved below for coverage checking.

▼ Summarize the occurrence of coffee. 4 Understanding Discuss Process flow for the manufacture of coffee Understanding

Summarize the occurrence of coffee. 4 Understanding Discuss Process flow for the manufacture of coffee Understanding is an official topic in Module 2 of Spices and Condiments. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify Summarize the occurrence of coffee. 4 Understanding Discuss Process flow for the manufacture of coffee Understanding using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, summarize the occurrence of coffee. 4 understanding discuss process flow for the manufacture of coffee understanding should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

▼ powder 5 Discuss Process flow for the manufacture of Understanding

powder 5 Discuss Process flow for the manufacture of Understanding is an official topic in Module 2 of Spices and Condiments. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify powder 5 Discuss Process flow for the manufacture of Understanding using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, powder 5 discuss process flow for the manufacture of understanding should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
--------	-----------------

Aspect	What to prepare
Definition/identity	What powder 5 Discuss Process flow for the manufacture of Understanding means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 2-പുറം powder 5 Discuss Process flow for the manufacture of Understanding പൊടി definition/meaning പൊടി. പൊടി പൊടി പൊടി, steps, application പൊടി പൊടി പൊടി. Technical terms English-പൊടി പൊടി പൊടി.

▼ instant powder 5 manufacturing process of coffee powder and instant coffee

instant powder 5 manufacturing process of coffee powder and instant coffee is an official topic in Module 2 of Spices and Condiments. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify instant powder 5 manufacturing process of coffee powder and instant coffee using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, instant powder 5 manufacturing process of coffee powder and instant coffee should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
--------	-----------------

Aspect	What to prepare
Definition/identity	What instant powder 5 manufacturing process of coffee powder and instant coffee means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 2- instant powder 5 manufacturing process of coffee powder and instant coffee definition/meaning . steps, application . Technical terms English- .

► Official syllabus detail and terminology (expand in digital version)

Module study routine: Read the source wording, make a one-page concept map, practise one short answer and one structured answer, then review the Malayalam support notes.

MODULE 3

Explain the occurrence, process flow, and manufacturing mechanisms of

This module is presented from the official syllabus scope for Spices and Condiments. Use the topic cards for learning and the source transcript for exact coverage.

As specified
official hours

CO-linked
see outcomes

Learning goals

- Explain the official Module 3 scope and its relationship to the course outcomes.
- Recognise the terminology, sequence, components and evidence expected in examination answers.
- Connect at least one official topic to a practical, industrial, laboratory or design context.

Key facts

- Official module title: Explain the occurrence, process flow, and manufacturing mechanisms of
- Official duration: As specified
- Official topic entries captured: 3
- The full source excerpt is preserved below for coverage checking.

▼ Discuss about Tea

Discuss about Tea is an official topic in Module 3 of Spices and Condiments. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify Discuss about Tea using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, discuss about tea should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What Discuss about Tea means in the official course context
Study lens	Principle → components or stages → result → application

▼ Explain Process flow for the manufacture of tea 5 Understanding Differentiate types of tea as white, green, oolong,

Explain Process flow for the manufacture of tea 5 Understanding Differentiate types of tea as white, green, oolong, is an official topic in Module 3 of Spices and Condiments. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify Explain Process flow for the manufacture of tea 5 Understanding Differentiate types of tea as white, green, oolong, using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, explain process flow for the manufacture of tea 5 understanding differentiate types of tea as white, green, oolong, should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What Explain Process flow for the manufacture of tea 5 Understanding Differentiate types of tea as white, green, oolong, means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 3- Explain Process flow for the manufacture of tea 5 Understanding Differentiate types of tea as white, green, oolong, definition/meaning . steps, application . Technical terms English- .

▼ 5 and black. Understanding manufacturing process of tea, different types of tea- white, green, oolong, black Describe the occurrence, processing mechanisms (primary and secondary),

5 and black. Understanding manufacturing process of tea, different types of tea- white, green, oolong, black Describe the occurrence, processing mechanisms (primary and secondary), is an official topic in Module 3 of Spices and Condiments. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify 5 and black. Understanding manufacturing process of tea, different types of tea- white, green, oolong, black Describe the occurrence, processing mechanisms (primary and secondary), using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, 5 and black. understanding manufacturing process of tea, different types of tea- white, green, oolong, black describe the occurrence, processing mechanisms (primary and secondary), should be discussed with

attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What 5 and black. Understanding manufacturing process of tea, different types of tea- white, green, oolong, black Describe the occurrence, processing mechanisms (primary and secondary), means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 3- 5 and black. Understanding manufacturing process of tea, different types of tea- white, green, oolong, black Describe the occurrence, processing mechanisms (primary and secondary), definition/meaning. steps, application. Technical terms English-

► Official syllabus detail and terminology (expand in digital version)

Module study routine: Read the source wording, make a one-page concept map, practise one short answer and one structured answer, then review the

Malayalam support notes.

MODULE 4

and quality control procedures involved in cocoa and chocolate production.

This module is presented from the official syllabus scope for Spices and Condiments. Use the topic cards for learning and the source transcript for exact coverage.

As specified
official hours

CO-linked
see outcomes

Learning goals

- Explain the official Module 4 scope and its relationship to the course outcomes.
- Recognise the terminology, sequence, components and evidence expected in examination answers.
- Connect at least one official topic to a practical, industrial, laboratory or design context.

Key facts

- Official module title: and quality control procedures involved in cocoa and chocolate production.
- Official duration: As specified
- Official topic entries captured: 3
- The full source excerpt is preserved below for coverage checking.

▼ of cocoa 6 Understanding Discuss about Process flow for cocoa processing

of cocoa 6 Understanding Discuss about Process flow for cocoa processing is an official topic in Module 4 of Spices and Condiments. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify of cocoa 6 Understanding Discuss about Process flow for cocoa processing using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, of cocoa 6 understanding discuss about process flow for cocoa processing should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
--------	-----------------

Aspect	What to prepare
Definition/identity	What of cocoa 6 Understanding Discuss about Process flow for cocoa processing means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 4- of cocoa 6 Understanding Discuss about Process flow for cocoa processing definition/meaning . , steps, application . Technical terms English- .

▼ into various products

into various products is an official topic in Module 4 of Spices and Condiments. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify into various products using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, into various products should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What into various products means in the official course context
Study lens	Principle → components or stages → result → application

▼ **Explain quality control of chocolates 6 Understanding cocoa processing - primary and secondary processing, Quality control of chocolates** Text / Reference T/R Book Title/Author Amalendu Chakraverty, Arun S. Mujumdar, G. S. Vijaya Raghavan & R1 Hosahalli S. Ramaswamy: “Handbook of Postharvest Technology: Cereals, Fruits, Vegetables, Tea, and Spices”, CRC Press, 2003. K.V. Peter, “Handbook of herbs & spices”, Volume1, ISBN: 0857095684, R2 9780857095688, Elsevier, 2012. N. Kumar, “Introduction to Spices, Plantation Crops, Medicinal and Aromatic R3 Plants”, Oxford & IBH Publishing Co Pvt. Ltd; 2nd Revised edition, ISBN-10: 8120417763, ISBN-13: 978-8120417762, 2018. Chi-Tang Ho, Jen-Kun Lin, Fereidoon Shahidi. “Tea and Tea Products: R4 Chemistry and Health-Promoting Properties, Nutraceutical Science and Technology”, CRC Press, 2008. Susheela Raghavan, “Handbook Seasoning, Spices & Flavoring”, Second Edition, R5 CRC Press, Publication, 2006. Online Resources Sl.No Website Link 1 <http://www.teabreeze.com/Manufacturingprocess.aspx> 2 <https://exclusivelane.com/blogs/handmade/a-brief-history-of-indian-spices> 3 <https://www.newfoodmagazine.com/article/28006/process-coffee-production-seed-cup/> <https://museum.dea.gov/exhibits/online-exhibits/cannabis-coca-and-poppy-natures-4-addictive-plants/coca>

Explain quality control of chocolates 6 Understanding cocoa processing - primary and secondary processing, Quality control of chocolates Text / Reference T/R Book Title/Author Amalendu Chakraverty, Arun S. Mujumdar, G. S. Vijaya Raghavan & R1 Hosahalli S. Ramaswamy: “Handbook of Postharvest Technology: Cereals, Fruits, Vegetables, Tea, and Spices”, CRC Press, 2003. K.V. Peter, “Handbook of herbs & spices”, Volume1, ISBN: 0857095684, R2 9780857095688, Elsevier, 2012. N. Kumar, “Introduction to Spices, Plantation Crops, Medicinal and Aromatic R3 Plants”, Oxford & IBH Publishing Co Pvt. Ltd; 2nd Revised edition, ISBN-10: 8120417763, ISBN-13: 978-8120417762, 2018. Chi-Tang Ho, Jen-Kun Lin, Fereidoon Shahidi. “Tea and Tea Products: R4 Chemistry and Health-Promoting Properties, Nutraceutical Science and Technology”, CRC Press, 2008. Susheela Raghavan, “Handbook Seasoning, Spices & Flavoring”, Second

Edition, R5 CRC Press, Publication, 2006. Online Resources Sl.No Website
 Link 1 <http://www.teabreeze.com/Manufacturingprocess.aspx> 2
<https://exclusivelane.com/blogs/handmade/a-brief-history-of-indian-spices>
 3 <https://www.newfoodmagazine.com/article/28006/process-coffee-production-seed-cup/> <https://museum.dea.gov/exhibits/online-exhibits/cannabis-coca-and-poppy-natures-4-addictive-plants/coca-is-an-official-topic-in-Module-4-of-Spices-and-Condiments>. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify Explain quality control of chocolates 6 Understanding cocoa processing – primary and secondary processing, Quality control of chocolates Text / Reference T/R Book Title/Author Amalendu Chakraverty, Arun S. Mujumdar, G. S. Vijaya Raghavan & R1 Hosahalli S. Ramaswamy: “Handbook of Postharvest Technology: Cereals, Fruits, Vegetables, Tea, and Spices”, CRC Press, 2003. K.V. Peter, “Handbook of herbs & spices”, Volume1, ISBN: 0857095684, R2 9780857095688, Elsevier, 2012. N. Kumar, “Introduction to Spices, Plantation Crops, Medicinal and Aromatic R3 Plants”, Oxford & IBH Publishing Co Pvt. Ltd; 2nd Revised edition, ISBN-10: 8120417763, ISBN-13: 978-8120417762, 2018. Chi-Tang Ho, Jen-Kun Lin, Fereidoon Shahidi. “Tea and Tea Products: R4 Chemistry and Health-Promoting Properties, Nutraceutical Science and Technology”, CRC Press, 2008. Susheela Raghavan, “Handbook Seasoning, Spices & Flavoring”, Second Edition, R5 CRC Press, Publication, 2006. Online Resources Sl.No Website Link 1 <http://www.teabreeze.com/Manufacturingprocess.aspx> 2 <https://exclusivelane.com/blogs/handmade/a-brief-history-of-indian-spices> 3 <https://www.newfoodmagazine.com/article/28006/process-coffee-production-seed-cup/> <https://museum.dea.gov/exhibits/online-exhibits/cannabis-coca-and-poppy-natures-4-addictive-plants/coca-is-an-official-topic-in-Module-4-of-Spices-and-Condiments>

exhibits/cannabis-coca-and-poppy-natures- 4 addictive-plants/coca using standard technical terminology.

- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, explain quality control of chocolates 6 understanding cocoa processing – primary and secondary processing, quality control of chocolates text / reference t/r book title/author amalendu chakraverty, arun s. mujumdar, g. s. vijaya raghavan & r1 hosahalli s. ramaswamy: “handbook of postharvest technology: cereals, fruits, vegetables, tea, and spices”, crc press, 2003. k.v. peter, “handbook of herbs & spices”, volume1, isbn: 0857095684, r2 9780857095688, elsevier, 2012. n. kumar, “introduction to spices, plantation crops, medicinal and aromatic r3 plants”, oxford & ibh publishing co pvt. ltd; 2nd revised edition, isbn-10: 8120417763, isbn-13: 978-8120417762, 2018. chi-tang ho, jen-kun lin, fereidoon shahidi. “tea and tea products: r4 chemistry and health-promoting properties, nutraceutical science and technology”, crc press, 2008. susheela raghavan, “handbook seasoning, spices & flavoring”, second edition, r5 crc press, publication, 2006. online resources sl.no website link 1 <http://www.teabreeze.com/manufacturingprocess.aspx> 2 <https://exclusivelane.com/blogs/handmade/a-brief-history-of-indian-spices> 3 <https://www.newfoodmagazine.com/article/28006/process-coffee-production-seed-cup/> <https://museum.dea.gov/exhibits/online-exhibits/cannabis-coca-and-poppy-natures-4-addictive-plants/coca> should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What Explain quality control of chocolates 6 Understanding cocoa processing – primary and secondary processing, Quality control of chocolates Text / Reference T/R Book Title/Author Amalendu

Aspect	What to prepare
	Chakraverty, Arun S. Mujumdar, G. S. Vijaya Raghavan & R1 Hosahalli S. Ramaswamy: "Handbook of Postharvest Technology: Cereals, Fruits, Vegetables, Tea, and Spices", CRC Press, 2003. K.V. Peter, "Handbook of herbs & spices", Volume1, ISBN: 0857095684, R2 9780857095688, Elsevier, 2012. N. Kumar, "Introduction to Spices, Plantation Crops, Medicinal and Aromatic R3 Plants", Oxford & IBH Publishing Co Pvt. Ltd; 2nd Revised edition, ISBN-10: 8120417763, ISBN-13: 978-8120417762, 2018. Chi-Tang Ho, Jen-Kun Lin, Fereidoon Shahidi. "Tea and Tea Products: R4 Chemistry and Health-Promoting Properties, Nutraceutical Science and Technology", CRC Press, 2008. Susheela Raghavan, "Handbook Seasoning, Spices & Flavoring", Second Edition, R5 CRC Press, Publication, 2006. Online Resources Sl.No Website Link 1 http://www.teabreeze.com/Manufacturingprocess.aspx 2 https://exclusivelane.com/blogs/handmade/a-brief-history-of-indian-spices 3 https://www.newfoodmagazine.com/article/28006/process-coffee-production-seed-cup/ https://museum.dea.gov/exhibits/online-exhibits/cannabis-coca-and-poppy-natures- 4 addictive-plants/coca means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 4-□□ Explain quality control of chocolates 6 Understanding cocoa processing – primary and secondary processing, Quality control of chocolates Text / Reference T/R Book Title/Author Amalendu Chakraverty, Arun S. Mujumdar, G. S. Vijaya Raghavan & R1 Hosahalli S. Ramaswamy: "Handbook of Postharvest Technology: Cereals, Fruits, Vegetables, Tea, and Spices", CRC Press, 2003. K.V. Peter, "Handbook

of herbs & spices”, Volume1, ISBN: 0857095684, R2 9780857095688, Elsevier, 2012. N. Kumar, “Introduction to Spices, Plantation Crops, Medicinal and Aromatic R3 Plants”, Oxford & IBH Publishing Co Pvt. Ltd; 2nd Revised edition, ISBN-10: 8120417763, ISBN-13: 978-8120417762, 2018. Chi-Tang Ho, Jen-Kun Lin, Fereidoon Shahidi. “Tea and Tea Products: R4 Chemistry and Health-Promoting Properties, Nutraceutical Science and Technology”, CRC Press, 2008. Susheela Raghavan, “Handbook Seasoning, Spices & Flavoring”, Second Edition, R5 CRC Press, Publication, 2006. Online Resources Sl.No Website Link 1 <http://www.teabreeze.com/Manufacturingprocess.aspx> 2 <https://exclusivelane.com/blogs/handmade/a-brief-history-of-indian-spices> 3 <https://www.newfoodmagazine.com/article/28006/process-coffee-production-seed-cup/> <https://museum.dea.gov/exhibits/online-exhibits/cannabis-coca-and-poppy-natures-4-addictive-plants/coca> <https://www.merriam-webster.com/dictionary/cocaine> definition/meaning <https://www.merriam-webster.com/dictionary/cocaine>. <https://www.merriam-webster.com/dictionary/cocaine> <https://www.merriam-webster.com/dictionary/cocaine>, steps, application <https://www.merriam-webster.com/dictionary/cocaine> <https://www.merriam-webster.com/dictionary/cocaine> <https://www.merriam-webster.com/dictionary/cocaine>. Technical terms English- <https://www.merriam-webster.com/dictionary/cocaine> <https://www.merriam-webster.com/dictionary/cocaine>.

► **Official syllabus detail and terminology (expand in digital version)**

Module study routine: Read the source wording, make a one-page concept map, practise one short answer and one structured answer, then review the Malayalam support notes.

RAPID REFERENCE**Concept and formula bank****Answer structure**

Definition → Principle →
Components/steps →
Application → Conclusion

Use this sequence for descriptive questions when the topic does not require a numerical formula.

Practical record

Aim → Apparatus → Procedure
→ Observation → Result →
Safety

Use this sequence for laboratory, workshop and field activities.

RAPID REVISION**Diagram and source-transcript library****Module 1 source and topic cards**

Jump to official terminology, topic explanations and study guidance.

Module 2 source and topic cards

Jump to official terminology, topic explanations and study guidance.

Module 3 source and topic cards

Jump to official terminology, topic explanations and study guidance.

Module 4 source and topic cards

Jump to official terminology, topic explanations and study guidance.

OFFICIAL SELF-LEARNING**Activities from the syllabus**

Use the extracted official activity wording as the record of required self-learning work.

The official PDF does not expose a separate activities heading in the extracted text; consult the source PDF pages for the exact activity list.

EXAMINATION PREPARATION

Assessment methodology

The following source extract preserves the assessment information detected in the official PDF. Verify mark conversions and institutional updates against the current source before an examination.

The official assessment structure is reproduced from the supplied PDF.

Practice-paper notice: The model paper below is generated for revision and is not an official examination paper.

ACTIVE RECALL

Module-wise questions and answers

These are practice questions based on official topic coverage, not official examination questions.

Module 1

► **Define or explain Explain Status and scope of spice and flavor 4 Understanding processing industries in India Discuss sources, production, selection criteria of. (3 marks)**

► **Define or explain Spices 3 Understanding Classify spices on the basis of origin, physical. (3 marks)**

► **Define or explain 3 Understanding characteristic sources, production, selection criteria, classification on the basis of origin, physical characteristic. (3 marks)**

Module 2

► **Define or explain Summarize the occurrence of coffee. 4 Understanding Discuss Process flow for the manufacture of coffee Understanding. (3 marks)**

► **Define or explain powder 5 Discuss Process flow for the manufacture of Understanding. (3 marks)**

► **Define or explain instant powder 5 manufacturing process of coffee powder and instant coffee. (3 marks)**

Module 3

► **Define or explain Discuss about Tea. (3 marks)**

► **Define or explain Explain Process flow for the manufacture of tea 5 Understanding Differentiate types of tea as white, green, oolong,. (3 marks)**

► **Define or explain 5 and black. Understanding manufacturing process of tea, different types of tea- white, green, oolong, black Describe the occurrence, processing mechanisms (primary and secondary),. (3 marks)**

Module 4

► **Define or explain of cocoa 6 Understanding Discuss about Process flow for cocoa processing. (3 marks)**

► **Define or explain into various products. (3 marks)**

► **Define or explain Explain quality control of chocolates 6 Understanding cocoa processing - primary and secondary processing, Quality control of chocolates Text / Reference T/R Book Title/Author Amalendu Chakraverty, Arun S. Mujumdar, G. S. Vijaya Raghavan & R1 Hosahalli S. Ramaswamy: "Handbook of Postharvest Technology: Cereals, Fruits, Vegetables, Tea, and Spices", CRC Press, 2003. K.V. Peter, "Handbook of herbs & spices", Volume1, ISBN: 0857095684, R2 9780857095688, Elsevier, 2012. N. Kumar, "Introduction to Spices, Plantation Crops, Medicinal and Aromatic R3 Plants", Oxford & IBH Publishing Co Pvt. Ltd; 2nd Revised edition, ISBN-10: 8120417763, ISBN-13: 978-8120417762, 2018. Chi-Tang Ho, Jen-Kun Lin, Fereidoon Shahidi. "Tea and Tea Products: R4 Chemistry and Health-Promoting Properties, Nutraceutical Science and Technology", CRC Press, 2008. Susheela Raghavan, "Handbook Seasoning, Spices & Flavoring", Second Edition, R5 CRC Press, Publication, 2006. Online Resources SI.No Website Link 1 <http://www.teabreeze.com/Manufacturingprocess.aspx> 2**

<https://exclusivelane.com/blogs/handmade/a-brief-history-of-indian-spices> 3 <https://www.newfoodmagazine.com/article/28006/process-coffee-production-seed-cup/> <https://museum.dea.gov/exhibits/online-exhibits/cannabis-coca-and-poppy-natures-> 4 [addictive-plants/coca](#). (3 marks)

PRACTICE ONLY

Practice Model Paper — Not an Official Examination Paper

Attempt one short definition, one structured explanation and one long answer from every module. Use the official assessment pattern reproduced above when selecting time and marks.

Module 1

1-mark definition: State the meaning of **Explain Status and scope of spice and flavor** 4 **Understanding processing industries in India** Discuss sources, production, selection criteria of.

3-mark explanation: Explain one principle, process or comparison from this module.

7-mark answer: Write a structured answer using a labelled diagram, table, procedure, example or application.

Module 2

1-mark definition: State the meaning of **Summarize the occurrence of coffee.** 4 **Understanding Discuss Process flow for the manufacture of coffee** Understanding.

3-mark explanation: Explain one principle, process or comparison from this module.

7-mark answer: Write a structured answer using a labelled diagram, table, procedure, example or application.

Module 3

1-mark definition: State the meaning of **Discuss about Tea.**

3-mark explanation: Explain one principle, process or comparison from

Module 4

1-mark definition: State the meaning of **of cocoa** 6 **Understanding Discuss about Process flow for cocoa processing.**

this module.

7-mark answer: Write a structured answer using a labelled diagram, table, procedure, example or application.

3-mark explanation: Explain one principle, process or comparison from this module.

7-mark answer: Write a structured answer using a labelled diagram, table, procedure, example or application.

QUICK REVISION

Exam checklist

Before writing

- Read the command word: define, explain, compare, draw, calculate, analyse or design.
- Use the exact course terminology from the source transcript.
- Plan labels, units, sequence and marks before writing.

Before submitting

- Check every module has been revised.
- Check diagrams, tables, formula symbols and units.
- Separate official syllabus information from personal elaboration.

REFERENCE VOCABULARY

Glossary

Study glossary

Term	Meaning in this lesson
Course Outcome	A capability the student should demonstrate after completing the course.
Module	An official syllabus unit with defined topics and hours.
CIE	Continuous Internal Evaluation.
ESE	End Semester Examination.
Source transcript	A preserved extract of the official syllabus used for coverage checking.

SOURCE-SUPPORTED REFERENCES

References and web resources

References listed in the official PDF

References are included in the official source PDF.

Web resources listed in the official PDF

- Sl.No Website Link
- <http://www.teabreeze.com/Manufacturingprocess.aspx>
- <https://exclusivelane.com/blogs/handmade/a-brief-history-of-indian-spices>
- <https://www.newfoodmagazine.com/article/28006/process-coffee-production-seed-cup/>
- <https://museum.dea.gov/exhibits/online-exhibits/cannabis-coca-and-poppy-natures-addictive-plants/coca>

IN-PAGE SEARCH**Search this lesson**

Prepared from the supplied official SITTTTR syllabus PDF for Course 5423B. Verify institutional updates against the current official curriculum.